Multilevel GRA Meta-Nullification Architecture in the Multiverse and All Types of GRA Nullification: Static, Cyclic, Chaotic, Hybrid

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Abstract

This paper presents a formal model of multilevel GRA Meta-Nullification in the multiverse. It defines state spaces, goal hierarchies, a recursive definition of foam, and a super-meta-functional. A theorem on complete multiverse nullification is proved. Furthermore, a classification of all types of GRA nullification according to the attractor type of a dynamical system is given: static (fixed point), cyclic (limit cycle), chaotic (strange attractor), and hybrid (combination). For each type, mathematical models, applicability criteria, and practical examples are provided. A multilevel algorithm and parallel optimization are described. The work has physical, cognitive, and philosophical interpretations, as well as connections to category theory.

1 Introduction

GRA Nullification (Geometric Recursive Analytics) is a family of methods for minimizing "foam" $\Phi^{(l)}$ (deviations from goals) in multilevel systems. Depending on the dynamical behavior of the system, three fundamental types of attractors are distinguished, corresponding to three types of GRA: static (fixed point), cyclic (limit cycle), and chaotic (strange attractor). In addition, hybrid variants (different hierarchical levels using different types) are possible.

The aim of this paper is to combine the general multilevel architecture of GRA Meta-Nullification in the multiverse with a detailed classification of all types of GRA, providing a unified mathematical framework for the analysis and optimization of complex systems — from physics and cosmology to AGI/ASI.

2 Multilevel GRA Meta-Nullification Architecture in the Multiverse

2.1 Extension to the Multiverse

GRA Multiverse is defined as an ordered or networked structure of many **meta-systems**, each of which is a complete GRA Meta-Nullification for its set of domains.

2.2 Formalization of Multiverse Levels

We have a hierarchy of meta-systems:

- **Level 0** (basic): local nullifications for individual domains.
- **Level 1** (meta): coordination of domains within a single meta-system.
- **Level 2** (meta-meta): coordination of different meta-systems.
- **Level K** (multiversal): complete coordination of the entire hierarchy.

Each object is indexed by a multi-index:

$$\mathbf{a} = (a_0, a_1, \ldots, a_k),$$

where a_0 is the domain index inside a meta-system, a_1 is the meta-system index inside a meta-meta-system, $\ldots$, a_k is the index at level k.

2.3 State Space of the Multiverse

For level l:

$$\mathcal{H}^{(l)} = \bigotimes_{\mathbf{a}:\dim(\mathbf{a})=l} \mathcal{H}^{(\mathbf{a})}.$$

The full multiverse space:

$$\mathcal{H}_{\text{multiverse}} = \bigotimes_{l=0}^{K} \mathcal{H}^{(l)}.$$

2.4 Goal Hierarchy and Projectors

Each level has its own goal:

- local goals $G_0^{(\mathbf{a})}$ for each domain;
- meta-goals $G_1^{(\mathbf{b})}$ for domain coordination;
- meta-meta goals $G_2^{(\mathbf{c})}$ for meta-system coordination;
- global multiverse goal G_K.

The projector onto the solution space of the goal at level l is denoted $\mathcal{P}_{G_l}$.

2.5 Recursive Definition of Foam

Level l foam for state $\Psi^{(l)}$ and goal G_l:

$$\Phi^{(l)}(\Psi^{(l)}, G_l) = \sum_{\substack{\mathbf{a}\neq\mathbf{b} \\ \dim(\mathbf{a})=\dim(\mathbf{b})=l}} \left|\langle\Psi^{(\mathbf{a})}|\mathcal{P}_{G_l}|\Psi^{(\mathbf{b})}\rangle\right|^2.$$

2.6 Super-Meta-Functional for the Multiverse

The total multiverse state is $\mathbf{\Psi} = \{\Psi^{(\mathbf{a})}\}_{\mathbf{a}\in\mathcal{I}}$. The functional:

$$J_{\text{multiverse}}(\mathbf{\Psi}) = \sum_{l=0}^{K} \Lambda_l \sum_{\substack{\mathbf{a} \\ \dim(\mathbf{a})=l}} J^{(l)}(\Psi^{(\mathbf{a})}),$$

where

$$J^{(0)}(\Psi^{(\mathbf{a})}) = J_{\text{loc}}(\Psi^{(\mathbf{a})}; G_0^{(\mathbf{a})}),$$
$$J^{(l)}(\Psi^{(\mathbf{a})}) = \sum_{\substack{\mathbf{b}\prec\mathbf{a} \\ \dim(\mathbf{b})=l-1}} J^{(l-1)}(\Psi^{(\mathbf{b})}) + \Phi^{(l)}(\Psi^{(\mathbf{a})}, G_l^{(\mathbf{a})}).$$

Here $\mathbf{b} \prec \mathbf{a}$ means $\mathbf{b}$ is a subsystem of $\mathbf{a}$. Hyperparameters:

$$\Lambda_l = \lambda_0 \cdot \alpha^l, \quad 0 < \alpha < 1.$$

2.7 Theorem of Complete Multiverse Nullification

Theorem (multiverse nullification). If the following conditions hold:

1. **Full commutativity**: $[\mathcal{P}_{G_l^{(\mathbf{a})}}, \mathcal{P}_{G_m^{(\mathbf{b})}}] = 0$ for all $\mathbf{a}, \mathbf{b}, l, m$.
2. **Hierarchy consistency**: $\mathcal{P}_{G_l^{(\mathbf{a})}} = \bigotimes_{\mathbf{b}\prec\mathbf{a}} \mathcal{P}_{G_{l-1}^{(\mathbf{b})}}$ for all $l \geq 1$.
3. **Space completeness**: $\dim(\mathcal{H}_{\text{multiverse}}) \geq \prod_{l=0}^{K} N_l$, where N_l is the number of subsystems at level l.

Then there exists a state $\mathbf{\Psi}^*$ such that

$$\Phi^{(l)}(\Psi^{(l)*}, G_l) = 0 \quad \forall l = 0, \ldots, K.$$

2.8 Multilevel Algorithm and Parallel Optimization

The algorithm is based on recursive level nullification and gradient descent. Parallel optimization uses the formula:

$$\frac{\partial J_{\text{multiverse}}}{\partial \Psi^{(\mathbf{a})}} = \Lambda_l \frac{\partial \Phi^{(l)}}{\partial \Psi^{(\mathbf{a})}} + \sum_{\mathbf{b}\succ\mathbf{a}} \Lambda_{l+1} \frac{\partial \Phi^{(l+1)}}{\partial \Psi^{(\mathbf{a})}}.$$

State update:

$$\Psi^{(\mathbf{a})}(t+1) = \Psi^{(\mathbf{a})}(t) - \eta \left[\Lambda_l \frac{\partial \Phi^{(l)}}{\partial \Psi^{(\mathbf{a})}} + \sum_{\mathbf{b}\succ\mathbf{a}} \Lambda_{l+1} \frac{\partial \Phi^{(l+1)}}{\partial \Psi^{(\mathbf{a})}} \right].$$

2.9 Algorithm Complexity

For K levels and N subsystems per level:

$$\text{Complexity} = O\left(\sum_{l=0}^{K} N^2 \alpha^l \right) = O\left(N^2 \frac{1-\alpha^{K+1}}{1-\alpha} \right).$$

As $K \to \infty$ and $\alpha < 1$, complexity remains $O(N^2/(1-\alpha))$ — polynomial.

3 All Types of GRA Nullification: Static, Cyclic, Chaotic, Hybrid

3.1 Static GRA (Fixed Point)

The system is described by $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}),\ \mathbf{x} \in \mathbb{R}^n$. Let $\mathbf{x}^*$ be a fixed point ($\mathbf{f}(\mathbf{x}^*) = 0$). Foam $\Phi(\mathbf{x}) = \|\mathbf{x} - \mathbf{x}_{\text{goal}}\|^2$.

Applicability criterion: all eigenvalues of the linearization matrix $A = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}\big|_{\mathbf{x}^*}$ satisfy $\mathrm{Re}(\lambda_i) < 0$ (asymptotic stability).

Examples: thermostat, robot position regulator, traveling salesman problem.

3.2 Cyclic GRA (Limit Cycle)

The system tends to a closed trajectory $\Gamma = \{\mathbf{x}(t) : \mathbf{x}(t+T) = \mathbf{x}(t)\}$ with minimal amplitude. Foam:

$$\Phi_{\text{cycle}} = \frac{1}{T}\int_0^T \|\mathbf{x}(t) - \mathbf{x}_{\text{ref}}(t)\|^2 dt.$$

Applicability criterion: at the fixed point, the linearization has a pair of purely imaginary eigenvalues $\pm i\omega$ (others $\mathrm{Re} < 0$) and the first Lyapunov coefficient $l_1 < 0$ (Andronov–Hopf bifurcation).

Examples: heartbeat, pendulum clock, Belousov–Zhabotinsky reaction, breathing.

3.3 Chaotic GRA (Strange Attractor)

The system exhibits deterministic chaos: trajectories are sensitive to initial conditions, unpredictable but bounded. Foam is the variance:

$$\Phi_{\text{chaos}} = \langle \|\mathbf{x} - \langle \mathbf{x} \rangle\|^2 \rangle.$$

Applicability criterion: largest Lyapunov exponent $\lambda_1 > 0$.

Examples: hurricanes, fluid turbulence, stock market crashes, cardiac fibrillation.

3.4 Hybrid GRA (Multilevel)

Different hierarchical levels use different GRA types. The total functional:

$$J_{\text{hybrid}} = \sum_{l=0}^{K} \Lambda_l \Phi^{(l)}_{\text{type}(l)}.$$

Applications: AGI/ASI (fast signal processing – static, planning – cyclic, creativity – chaos); robots (posture stabilization, walking, environmental adaptation); ecosystems (population cycles, climate fluctuations, reserve management).

4 Comparative Table

5 Philosophical and Cognitive Interpretations

The GRA multiverse implements **transcendental objectivity**, eliminating subjectivity, domain specificity, and hierarchical artifacts. The state $\Psi^*_{\text{multiverse}}$ is an **absolute cognitive vacuum** — complete knowledge without interpretational layers. In the limit of

Table 1: Types of GRA Nullification

GRA Type	Attractor	Criterion	Example
Static	Fixed point	$\mathrm{Re}(\lambda_i) < 0$	Thermostat
Cyclic	Limit cycle	$\lambda_{1,2} = \pm i\omega,\ l_1 < 0$	Heartbeat
Chaotic	Strange attractor	$\lambda_1 > 0$	Hurricane
Hybrid	Mixed	Time scale hierarchy	AGI/ASI

infinite hierarchy ($K \to \infty$), we obtain a fixed point:

$$\boxed{\lim_{K\to\infty} \Psi_K^* = \Psi_\infty^*, \quad \bigcap_{l=0}^{\infty} \ker(\Phi^{(l)}) = \{\Psi_\infty^*\}}.$$

6 Conclusion

The unified architecture of GRA Meta-Nullification in the multiverse, together with the classification of all types of GRA, provides a powerful mathematical tool for modeling, optimizing and understanding complex systems — from physical to cognitive and social. Convergence to a state of complete foam nullification, achievable in polynomial time, is proved.

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